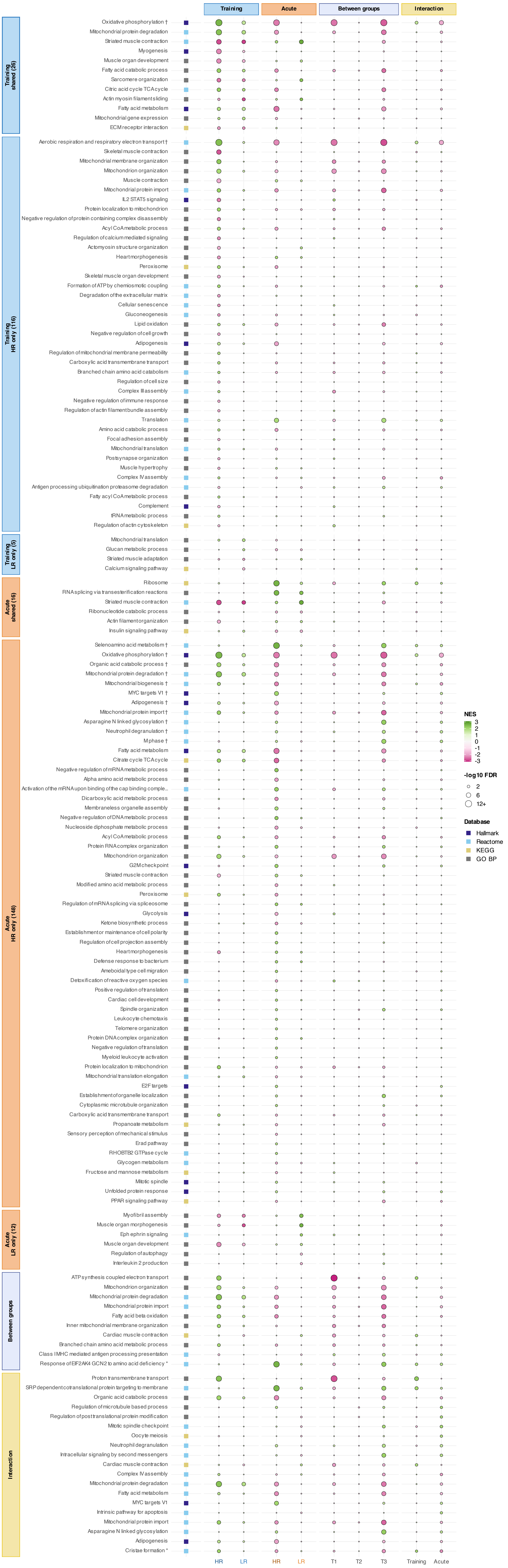
a **Every candidate set, grouped by tree**

Sets in leaf order of each block's tree (shared proteins, Jaccard, average linkage); bands alternate between clusters at height 0.85
bold: the cluster's most specific set; * calls of opposite sign; † interaction FDR < 0.05

Training
Acute
Between groups
Interaction
Training shared (26)
ECM receptor interaction
ECM proteoglycans
Actin myosin filament sliding
Muscle organ morphogenesis
Cardiac muscle tissue morphogenesis
Myogenesis
Muscle organ development
Muscle cell development
Sarcomere organization
Myofibril assembly
Cellular component assembly involved in morphogenesis
Striated muscle contraction
Muscle contraction
Mitochondrial gene expression
Fatty acid metabolism
Organic acid catabolic process
Monocarboxylic acid catabolic process
Fatty acid catabolic process
Fatty acid beta oxidation
Citric acid cycle TCA cycle
Tricarboxylic acid cycle
Citrate cycle TCA cycle
Mitochondrial protein degradation
Oxidative phosphorylation †
Cellular respiration
Generation of precursor metabolites and energy
Training HR only (116)
Carboxylic acid transmembrane transport
Peroxisome
Peroxisome
Adipogenesis
Fatty acyl CoA metabolic process
Fatty acid derivative metabolic process
Amide biosynthetic process
Acyl CoA metabolic process
Acyl CoA biosynthetic process
Lipid catabolic process
Lipid modification
Lipid oxidation
Fatty acid metabolic process
Fatty acid metabolism
Valine leucine and isoleucine degradation
Branched chain amino acid metabolic process
Branched chain amino acid catabolism
Small molecule catabolic process
Amino acid catabolic process
Amino acid metabolic process
tRNA metabolic process
Antigen processing ubiquitination proteasome degradation
Translation
Mitochondrial translation
Apoptotic mitochondrial changes
Mitochondrial transport
Regulation of mitochondrial membrane permeability
Regulation of membrane permeability
Mitochondrial membrane organization
Inner mitochondrial membrane organization
Mitochondrial protein import
Protein localization
Protein transmembrane transport
Protein localization to mitochondrion
Establishment of protein localization to mitochondrion
Complex III assembly
Complex IV assembly
Organelle biogenesis and maintenance
Formation of ATP by chemiosmotic coupling
Cristae formation †
Mitochondrial biogenesis
ATP metabolic process
Purine containing compound biosynthetic process
Nucleoside phosphate biosynthetic process
Nucleotide biosynthetic process
Ribose phosphate biosynthetic process
Proton motive force driven ATP synthesis †
ATP biosynthetic process
Nucleoside triphosphate biosynthetic process
Mitochondrial electron transport NADH to ubiquinone †
Complex I biogenesis †
Aerobic respiration and respiratory electron transport †
Respiratory electron transport †
Oxidative phosphorylation †
Electron transport chain †
ATP synthesis coupled electron transport †
Proton transmembrane transport †
Oxidative phosphorylation †
Mitochondrion organization
Mitochondrial respiratory chain complex assembly
Gluconeogenesis
Monosaccharide catabolic process
IL2 STAT5 signaling
Degradation of the extracellular matrix
Calcium mediated signaling
Regulation of calcium mediated signaling
Heart morphogenesis
Actomyosin structure organization
Muscle cell differentiation
Striated muscle cell differentiation
Skeletal muscle organ development
Muscle tissue development
Cardiac muscle tissue development
Muscle hypertrophy
Muscle adaptation
Neuromuscular process
Multicellular organismal movement
Skeletal muscle contraction
Muscle contraction
Striated muscle contraction
Regulation of muscle system process
Regulation of muscle contraction
Cardiac muscle contraction
Heart process
Regulation of blood circulation
Regulation of heart contraction
Negative regulation of growth
Negative regulation of cell growth
Regulation of cell size
Regulation of developmental growth
Actin filament organization
Regulation of cellular component size
Regulation of actin filament length
Regulation of supramolecular fiber organization
Regulation of actin filament based process
Regulation of actin filament organization
Protein containing complex disassembly
Negative regulation of protein containing complex assembly
Negative regulation of cytoskeleton organization
Negative regulation of supramolecular fiber organization
Negative regulation of protein containing complex disassembly
Negative regulation of protein polymerization
Regulation of protein containing complex disassembly
Actin filament depolymerization
Regulation of protein depolymerization
Protein depolymerization
Postsynapse organization
Focal adhesion
Regulation of actin cytoskeleton
Regulation of actin filament bundle assembly
Cell substrate junction organization
Focal adhesion assembly
Cellular senescence
Toll like receptor TLR1 TLR2 cascade
Negative regulation of immune response
Complement
Training LR only (5)
Mitochondrial translation
Glucan metabolic process
Polysaccharide metabolic process
Striated muscle adaptation
Calcium signaling pathway
Acute shared (16)
RNA splicing via transesterification reactions
RNA splicing
Spliceosome
mRNA 3 end processing
mRNA polyadenylation
Processing of capped intron containing pre mRNA
mRNA splicing
Ribonucleotide catabolic process
Insulin signaling pathway
Ribosome
SRP dependent cotranslational protein targeting to membrane
Myotube differentiation
Muscle cell differentiation
Striated muscle cell differentiation
Striated muscle contraction
Actin filament organization
Acute HR only (148)
Sensory perception of mechanical stimulus
Modified amino acid metabolic process
Peroxisome
Detoxification of reactive oxygen species
Ketone biosynthetic process
Ketone metabolic process
Carboxylic acid transmembrane transport
PPAR signaling pathway
Dicarboxylic acid metabolic process
Citric acid cycle TCA cycle
Citrate cycle TCA cycle
Tricarboxylic acid cycle
Sulfur compound metabolic process
Acyl CoA metabolic process
Acetyl CoA metabolic process
Acyl CoA biosynthetic process
Propanoate metabolism
Pyruvate metabolism
Adipogenesis †
Branched chain amino acid metabolic process
Valine leucine and isoleucine degradation
Amino acid metabolic process
Alpha amino acid metabolic process
L amino acid metabolic process
Amino acid catabolic process
L amino acid catabolic process
Fatty acid metabolism
Mitochondrial fatty acid beta oxidation
Fatty acid metabolism †
Fatty acid metabolic process
Fatty acid metabolism
Small molecule catabolic process †
Organic acid catabolic process †
Lipid catabolic process
Monocarboxylic acid catabolic process
Lipid oxidation
Lipid modification
Fatty acid beta oxidation
Fatty acid catabolic process
Formation of ATP by chemiosmotic coupling †
Mitochondrial biogenesis †
Cristae formation †
Mitochondrion organization
Protein localization to mitochondrion
Mitochondrial protein import †
Protein localization
Glycogen metabolism
Glucan metabolic process
Polysaccharide metabolic process
Mitochondrial protein degradation †
Ribose phosphate metabolic process
ATP metabolic process
Nucleoside triphosphate metabolic process
Oxidative phosphorylation †
Generation of precursor metabolites and energy
Cellular respiration
Aerobic respiration and respiratory electron transport †
Respiratory electron transport †
Glycolysis
Fructose and mannose metabolism
Monosaccharide metabolic process
Glucose metabolic process
Pyridine containing compound metabolic process
Carbohydrate derivative catabolic process
Organophosphate catabolic process
Nucleoside phosphate catabolic process
Purine containing compound catabolic process
Carbohydrate catabolic process
Pyruvate metabolic process
Nucleoside diphosphate metabolic process
Nucleoside diphosphate catabolic process
Pyridine containing compound catabolic process
Glycolysis gluconeogenesis
Metabolism of carbohydrates and carbohydrate derivatives
Glucose metabolism
Mitochondrial gene expression
Mitochondrial translation
Mitochondrial translation elongation
Erad pathway
Unfolded protein response
Ribosome biogenesis
rRNA metabolic process
rRNA processing
Activation of the mRNA upon binding of the cap binding comple…
Ribosomal small subunit biogenesis
Striated muscle contraction
Muscle contraction
Myoblast fusion
Syncytium formation
Negative regulation of syncytium formation by plasma membrane…
Regulation of syncytium formation by plasma membrane fusion
Ribosome quality control RQC complex extracts and degrades na… †
Signaling by robo receptors †
Ribosome associated quality control †
Regulation of expression of slits and robos †
Selenoamino acid metabolism †
Cellular response to starvation †
Nonsense mediated decay NMD †
Eukaryotic translation elongation †
rRNA processing †
Response of EIF2AK4 GCN2 to amino acid deficiency †
Cytoplasmic translation †
Eukaryotic translation initiation †
Translation †
Metabolism of amino acids and derivatives
MYC targets V1 †
Protein RNA complex organization
Translational initiation
G2M checkpoint
E2F targets
Regulation of RNA splicing
Regulation of mRNA splicing via spliceosome
Regulation of mRNA processing
Nuclear transcribed mRNA catabolic process
Negative regulation of mRNA metabolic process
Negative regulation of mRNA catabolic process
Negative regulation of RNA catabolic process
Regulation of mRNA metabolic process
Regulation of mRNA catabolic process
mRNA catabolic process
Negative regulation of translation
Regulation of translation
Positive regulation of translation
RHOBTB2 GTPase cycle
Regulation of chromosome organization
Telomere organization
Negative regulation of DNA metabolic process
Protein DNA complex organization
Leukocyte chemotaxis
Defense response to bacterium
Myeloid leukocyte activation
Neutrophil degranulation †
Asparagine N linked glycosylation †
M phase †
Heart morphogenesis
Cardiac muscle tissue development
Cardiac cell development
Cardiocyte differentiation
Cardiac muscle cell differentiation
Regulation of cell projection assembly
Establishment or maintenance of cell polarity
Ameboidal type cell migration
Cytoplasmic microtubule organization
Mitotic spindle
Establishment of organelle localization
Membraneless organelle assembly
Regulation of microtubule based process †
Spindle organization
Acute LR only (12)
Eph ephrin signaling
Actin myosin filament sliding
Muscle organ morphogenesis
Cardiac muscle tissue morphogenesis
Muscle organ development
Muscle cell development
Actomyosin structure organization
Sarcomere organization
Myofibril assembly
Cellular component assembly involved in morphogenesis
Regulation of autophagy
Interleukin 2 production
Between groups
Class I MHC mediated antigen processing presentation
Translation *
Activation of the mRNA upon binding of the cap binding comple… *
SRP dependent cotranslational protein targeting to membrane *
Nonsense mediated decay NMD *
Eukaryotic translation elongation *
rRNA processing *
Response of EIF2AK4 GCN2 to amino acid deficiency *
Ribosome *
Eukaryotic translation initiation *
Cytoplasmic translation *
Lipid modification
Fatty acid catabolic process
Fatty acid beta oxidation
Valine leucine and isoleucine degradation
Branched chain amino acid metabolic process
Branched chain amino acid catabolism
Mitochondrial protein import
Protein localization
Protein localization to mitochondrion
Mitochondrial membrane organization
Inner mitochondrial membrane organization
Cardiac muscle contraction
Mitochondrion organization
Mitochondrial respiratory chain complex assembly
Mitochondrial protein degradation
Oxidative phosphorylation
Cellular respiration
Generation of precursor metabolites and energy
Ribose phosphate metabolic process
Nucleoside triphosphate metabolic process
ATP metabolic process
Proton motive force driven ATP synthesis
ATP biosynthetic process
Nucleoside triphosphate biosynthetic process
Nucleotide biosynthetic process
Purine containing compound biosynthetic process
Nucleoside phosphate biosynthetic process
Complex I biogenesis
Mitochondrial electron transport NADH to ubiquinone
Aerobic respiration and respiratory electron transport
Respiratory electron transport
Oxidative phosphorylation
ATP synthesis coupled electron transport
Electron transport chain
Proton transmembrane transport
Oxidative phosphorylation
Interaction
Adipogenesis
Fatty acid metabolism
Organic acid catabolic process
Small molecule catabolic process
Cardiac muscle contraction
Complex IV assembly
Mitochondrial protein degradation
Oxidative phosphorylation *
Proton motive force driven ATP synthesis
Mitochondrial electron transport NADH to ubiquinone
Complex I biogenesis
Aerobic respiration and respiratory electron transport *
Respiratory electron transport *
Oxidative phosphorylation
ATP synthesis coupled electron transport
Electron transport chain
Proton transmembrane transport
Oxidative phosphorylation
Mitochondrial protein import
Formation of ATP by chemiosmotic coupling
Mitochondrial biogenesis
Cristae formation *
Regulation of post translational protein modification
MYC targets V1
Translation
Ribosome quality control RQC complex extracts and degrades na…
Signaling by robo receptors
Ribosome associated quality control
Regulation of expression of slits and robos
Selenoamino acid metabolism
Cellular response to starvation
SRP dependent cotranslational protein targeting to membrane
Nonsense mediated decay NMD
Eukaryotic translation elongation
Response of EIF2AK4 GCN2 to amino acid deficiency
Ribosome
rRNA processing
Cytoplasmic translation
Eukaryotic translation initiation
Neutrophil degranulation
Regulation of microtubule based process
Oocyte meiosis
Intrinsic pathway for apoptosis
Apoptosis
Programmed cell death
Mitotic G2 G2 M phases
Cell cycle checkpoints
M phase
Mitotic metaphase and anaphase
Separation of sister chromatids
MAPK family signaling cascades
MAPK1 MAPK3 signaling
Intracellular signaling by second messengers
PTEN regulation
Asparagine N linked glycosylation
Mitotic spindle checkpoint
Mitotic prometaphase
Resolution of sister chromatid cohesion
Cilium assembly
Regulation of PLK1 activity at G2 M transition
Recruitment of NUMA to mitotic centrosomes
Anchoring of the basal body to the plasma membrane
AURKA activation by TPX2
Recruitment of mitotic centrosome proteins and complexes
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP

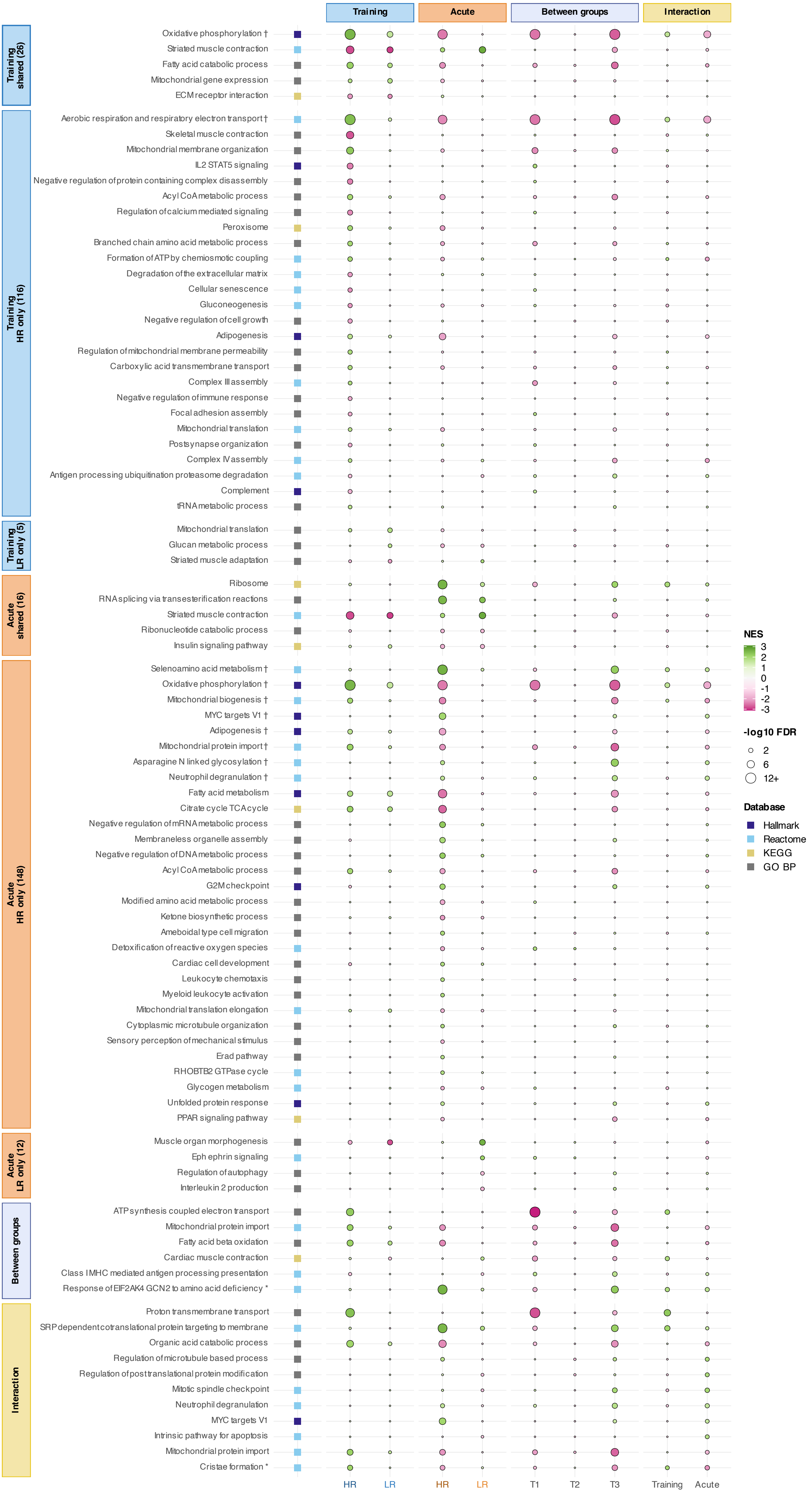
b One set per cluster at tree height 0.75

As Figure 3 with the tree cut at 0.75 instead of 0.85, every cluster shown; * calls of opposite sign; † interaction FDR < 0.05



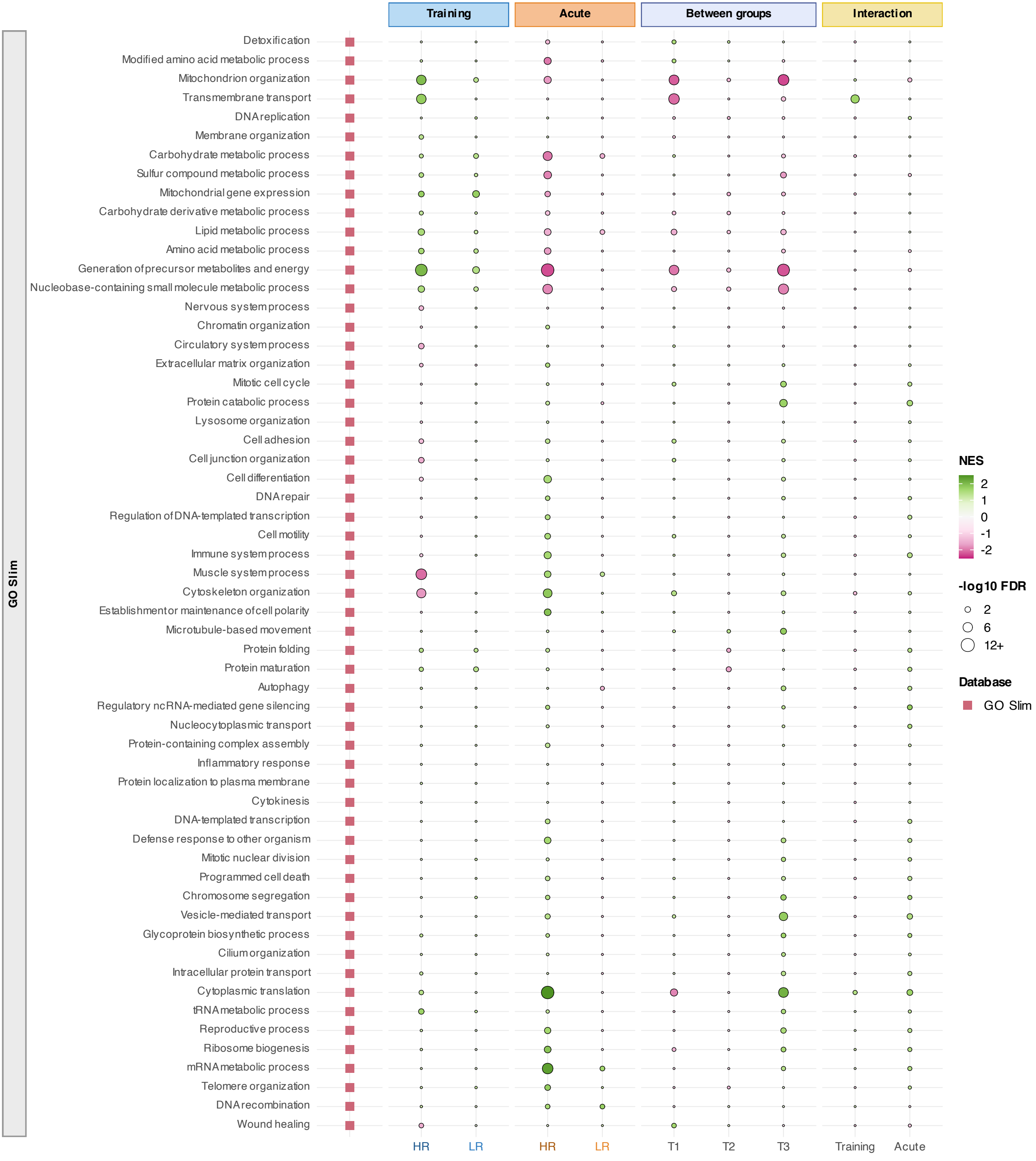
c **One set per cluster at tree height 0.9**

As Figure 3 with the tree cut at 0.9 instead of 0.85, every cluster shown; * calls of opposite sign; † interaction FDR < 0.05



d GO Slim themes in the nine contrasts

Every GO Slim theme; rows ordered by average-linkage clustering of their NES


Training
Acute
Between groups
Interaction
GO Slim
Detoxification
Modified amino acid metabolic process
Mitochondrion organization
Transmembrane transport
DNA replication
Membrane organization
Carbohydrate metabolic process
Sulfur compound metabolic process
Mitochondrial gene expression
Carbohydrate derivative metabolic process
Lipid metabolic process
Amino acid metabolic process
Generation of precursor metabolites and energy
Nucleobase-containing small molecule metabolic process
Nervous system process
Chromatin organization
Circulatory system process
Extracellular matrix organization
Mitotic cell cycle
Protein catabolic process
Lysosome organization
Cell adhesion
Cell junction organization
Cell differentiation
DNA repair
Regulation of DNA-templated transcription
Cell motility
Immune system process
Muscle system process
Cytoskeleton organization
Establishment or maintenance of cell polarity
Microtubule-based movement
Protein folding
Protein maturation
Autophagy
Regulatory ncRNA-mediated gene silencing
Nucleocytoplasmic transport
Protein-containing complex assembly
Inflammatory response
Protein localization to plasma membrane
Cytokinesis
DNA-templated transcription
Defense response to other organism
Mitotic nuclear division
Programmed cell death
Chromosome segregation
Vesicle-mediated transport
Glycoprotein biosynthetic process
Cilium organization
Intracellular protein transport
Cytoplasmic translation
tRNA metabolic process
Reproductive process
Ribosome biogenesis
mRNA metabolic process
Telomere organization
DNA recombination
Wound healing
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
2
1
0
-1
-2
-log10 FDR
2
6
12+
Database
GO Slim